

recordings that could ultimately be used to train commercial AI systems.

Another significant challenge is the lack of scalable, general-purpose models that can operate across different robot types and task categories. Physical environments require a high degree of situational awareness and contextual understanding to solve complex decision-making problems.

The cost and infrastructure requirements associated with physical AI present another major obstacle. Because real-world data must be collected, simulation environments developed, and expensive custom hardware deployed for testing and validation. Training and deploying physical AI systems is significantly more expensive than developing traditional software-based AI systems. While the gap between simulated and real-world performance continues to narrow, companies must still invest heavily in development, deployment, and physical experimentation.

The challenge of integrating VLMs and VLA models with robotic control systems also remains substantial. Vision-language models are highly effective at understanding text and visual information, but translating that understanding into meaningful physical actions remains an early-stage capability.

## Robotics startups: The reality behind the hype

One of the first and most fundamental challenges is market validation. The most valuable technologies are often those that solve problems users genuinely care about, regardless of whether those users understand the underlying technology. Users ultimately care about outcomes: whether a solution delivers results that are better, faster, or more cost-effective than existing alternatives. This creates strong demand for technologies that generate measurable near-term value rather than technical

## The physical AI stack

1. **Perception layer (sensing the physical world).** This is where physical AI first interacts with the real world using sensors like cameras, IMUs, and force sensors. These devices capture visual data, motion, and environmental signals. For example, wearable cameras on a worker's head or hands record how tasks are performed in real time.
2. **Data collection and representation layer.** Raw sensor data is cleaned and structured into usable formats. Videos are broken into action sequences, and motion data is converted into trajectories or key events.
3. **Learning layer (model training/imitation learning).** AI models learn from the prepared data using methods like imitation learning or reinforcement learning.
4. **Decision-making layer (policy generation).** This layer decides what the system should do next. It converts learned knowledge into actionable policies that map situations to actions.
5. **Control layer (robot/device control).** Here, high-level decisions are translated into low-level commands like motor speeds, joint angles, or force adjustments.
6. **Actuation layer (physical execution).** This is the final output stage where actions happen in the real world. Robots or machines perform tasks like moving objects, assembling parts, or interacting with environments.

sophistication alone.

Another major constraint is patient capital. Unlike software or purely digital AI companies, robotics ventures are capital-intensive and slow to iterate. Reaching a commercial product requires multiple design cycles, hardware revisions, physical testing, and field validation. In more mature innovation ecosystems, investors are often willing to commit significant capital over long time horizons before expecting returns. In India, while funding is available, patient capital for deep-tech hardware ventures remains comparatively limited.

Supply-chain complexity presents another challenge for deep-tech industries. Many critical robotics components are still not manufactured domestically and remain heavily dependent on overseas suppliers, particularly within China's established manufacturing ecosystem.

These supply-chain gaps also create opportunities for entrepreneurship, particularly in hardware manufacturing, component production, and specialised industrial services. At the entrepreneurial level, most robotics-focused deep-tech ventures require multidisciplinary teams. While building teams for conventional software ventures can be relatively straightforward,

robotics companies often require expertise spanning multiple domains.

Government agencies and other stakeholders have undertaken several promising initiatives to support technological entrepreneurship. Grants, incubators, and research programmes have been particularly helpful during the prototype-development stage. However, these programmes often provide limited support for commercialisation, leaving founders responsible for securing additional funding to scale production and market adoption.

India is still developing integrated manufacturing and prototyping ecosystems comparable to China's industrial clusters. Research parks and innovation hubs, particularly in Bengaluru, aim to bring suppliers, startups, and research institutions closer together to accelerate development.

The robotics and AI sector will continue advancing through both technological breakthroughs and expanding real-world applications. Wider adoption is expected over the next two to five years as systems demonstrate greater reliability, safety, and practical value. **EFY**

Co-author Nidhi Agarwal is Senior Technology Journalist at EFY with a deep interest in embedded systems, development boards, and IoT cloud solutions.